Modeling Bimanual Perceptual Interactions in Macaque Monkeys

Bayesian Hierarchical Modeling

Vera Schroeder

October 29, 2026

Outline

Task and Data

Model A: Sessions Pooled
- Model Structure
- Synthetic Data
- Real Data

Model B: Sessions Hierarchical
- Model Structure
- Synthetic Data
- Real Data

Model C: Distractor Amplitude as a Covariate

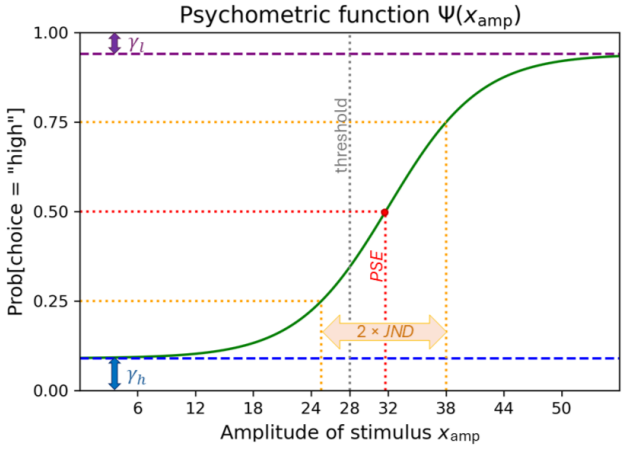
Summary

The task: 2AFC categorization of vibration amplitude

- stimulus $x_{\text{stim}} \in \{6, 12, 18, 24, 32, 38, 44, 50\}\mu m$
  - administered to one finger of one hand (left or right)
- distractor $x_{\text{dist}} \in \{0, x_{\text{d}}\}\mu m$
  - $x_{\text{d}} \in \{6, 12, 18, 24, 32\}\mu m$ value fixed for each session
  - administered to the other (non-stimulus) hand
- if present, distractor administered simultaneously with the stimulus
- monkey categorizes the stimulus as either "high" or "low" relative to a trained boundary of 28 μm.
  - "low" response: correct for $x_{\text{stim}} < 28\mu m$
  - "high" response: correct for $x_{\text{stim}} > 28\mu m$.
- correct responses are rewarded with water or juice

Psychometric function: response $r \sim \text{Bernoulli}(p = \psi)$

$$\psi(x_{\text{stim}}) = \gamma_h + (1 - \gamma_h - \gamma_l)\ \text{logit}^{-1}(\beta_0 + \beta_1 x_{\text{stim}})$$

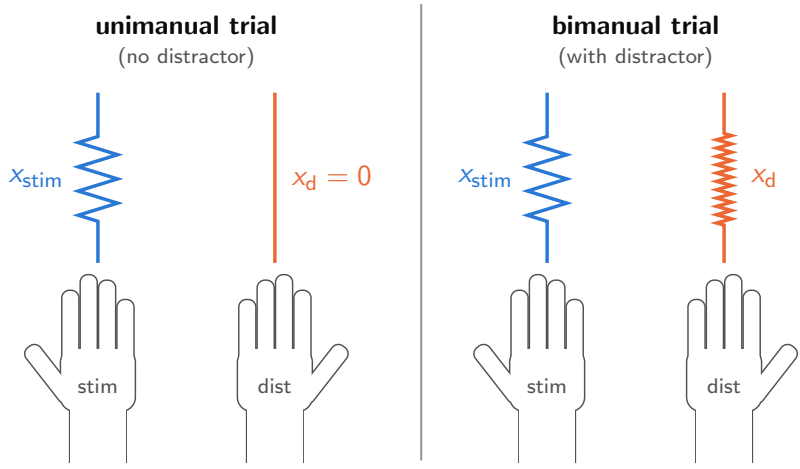


Parameters

$$\psi(x_{\text{stim}}) = \gamma_h + (1 - \gamma_h - \gamma_l)\ \text{logit}^{-1}(\beta_0 + \beta_1 x_{\text{stim}}), \qquad \vec{\theta} = (\beta_0, \beta_1, \gamma_h, \gamma_l)$$

- γ_h: lapse rate for "high"
  - lower bound on $\mathbf{P}(\text{guess} = \text{"high"})$, even to the weakest stimuli
  - lower (left) asymptote of ψ
- γ_l: lapse rate for "low"
  - lower bound on $\mathbf{P}(\text{guess} = \text{"low"})$, even to the strongest stimuli
  - $1 - \gamma_l$ = upper (right) asymptote of ψ
- logit^{-1}: the logistic sigmoid, $\text{logit}^{-1}(z) = \dfrac{1}{1 + \exp(-z)}$
- PSE: stimulus where responses of "high" and "low" are equally likely
  - x_{stim} such that $\psi(x_{\text{stim}}) = 0.5$
  - represents bias
- JND: how much the stimulus needs to change to be noticed half the time
  - half the distance between $\psi(x_{\text{stim}}) = 0.25$ and 0.75
  - represents sensitivity

The task: one trial



Modeling bimanual interactions

- stimulus $x_{\text{stim}} \in \{6, 12, 18, 24, 32, 38, 44, 50\}\mu m$
  - administered to one finger of one hand (left or right)
- distractor $x_{\text{dist}} \in \{0, x_{\text{d}}\}\mu m$
  - $x_{\text{d}} \in \{6, 12, 18, 24, 32\}\mu m$ value fixed for each session
  - administered to the other (non-stimulus) hand
- if present, distractor administered simultaneously with the stimulus
- monkey categorizes the stimulus as either "high" or "low" relative to a trained boundary of 28 μm.
  - "low" response: correct for $x_{\text{stim}} < 28\mu m$
  - "high" response: correct for $x_{\text{stim}} > 28\mu m$.
- correct responses are rewarded with water or juice

The data

- One monkey (Sirius), 12,937 trials.
- Collected over 42 sessions, averaging approximately 308 trials per session.
- 8 stimulus amplitudes, 6–50 μm; 4 groups: hand $\times$ (unimanual, bimanual).
- Each session uses a single distractor amplitude (6–32 μm) on its bimanual trials.

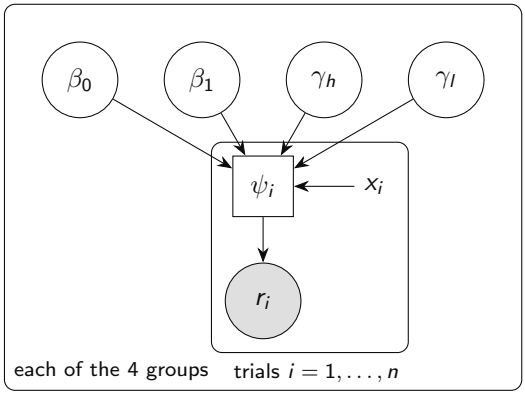
	left hand		right hand		
x_{stim} (μm)	unimanual	bimanual	unimanual	bimanual	total
6	408	404	404	402	1,618
12	407	400	408	400	1,615
18	402	400	403	405	1,610
24	402	405	402	406	1,615
32	404	405	405	407	1,621
38	403	408	404	408	1,623
44	406	403	404	405	1,618
50	405	405	406	401	1,617
total	3,237	3,230	3,236	3,234	12,937

Three models

Model	Sessions	Distractor
A	pooled: one curve per group	uni and bi fit separately
B	hierarchical: one curve per group and session	uni and bi fit separately
C	hierarchical, one group per hand	covariate: shift $\beta_2 d$

For each model: convergence study on synthetic data, then the fit to the real data.

Model A: graphical model



- circles: random variables
- shaded: observed
- rectangles: deterministic
- x_i: fixed covariate
- outer plate: the same model, fit independently to each group

Model A: likelihood

For each group (left/right $\times$ unimanual/bimanual), fit independently to its $n \approx 3{,}230$ trials from all 42 sessions pooled:

Likelihood: probability of obtaining responses $\vec{r}$ given parameters $\vec{\theta} = (\beta_0, \beta_1, \gamma_h, \gamma_l)$
$r_i \sim \textbf{Bernoulli}[\psi(x_i; \vec{\theta})]$ ($r_i = 1$ for "high", x_i standardized), so:

$$\mathbf{P}[r_i = z] = \psi(x_i; \vec{\theta})^z \left(1 - \psi(x_i; \vec{\theta})\right)^{1-z}, \qquad z \in \{0, 1\}$$

$$\begin{aligned} L(\vec{\theta}; \vec{r}) &= \prod_{i=1}^{n} \mathbf{P}[r_i \mid \vec{\theta}] \\ &= \prod_{i=1}^{n} \psi(x_i; \vec{\theta})^{r_i} \left(1 - \psi(x_i; \vec{\theta})\right)^{1-r_i} \end{aligned}$$

Model A: priors

The same priors in every group:

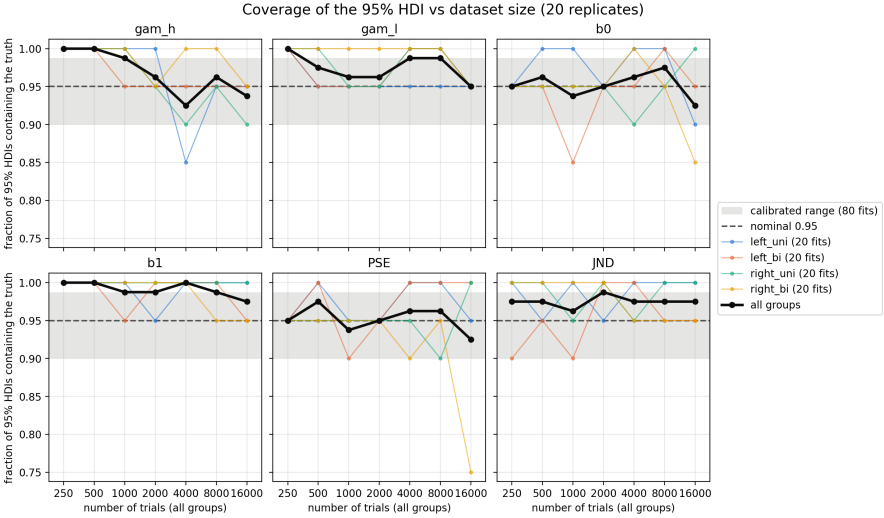
$$\begin{aligned}
\beta_0 &\sim \mathcal{N}(0,\ 6) \\
\beta_1 &\sim \mathcal{N}(4,\ 2) \\
\gamma_h &= 0.25\, w_h, \qquad & w_h &\sim \text{Beta}(1.22,\ 4.89) \\
\gamma_l &= 0.25\, w_l, \qquad & w_l &\sim \text{Beta}(1.22,\ 4.89)
\end{aligned}$$

- β_1 must be positive for the psychometric function to have a positive slope
- γ's have support between 0 and 0.25, with prior mean of 0.05 and 95% range [0.003, 0.141]

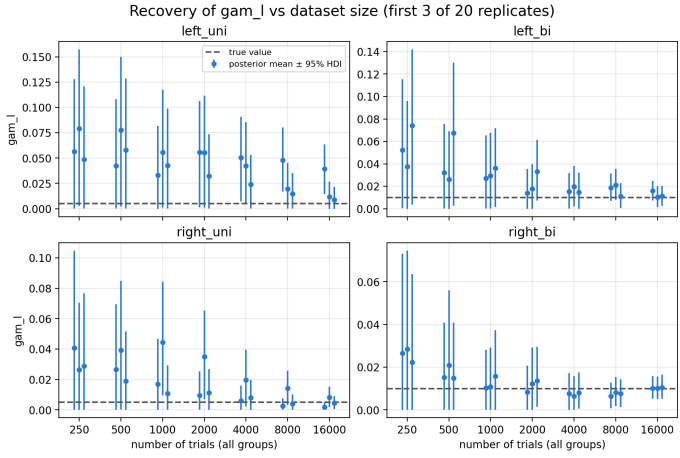
Model A on synthetic data: the study

- One fixed set of true parameters $(\beta_0, \beta_1, \gamma_h, \gamma_l)$ per group
- Trials drawn from the real design (the real mix of stimulus levels and groups), responses simulated from the true curves
- 7 sizes: 250 to 16,000 trials; the real data (12,937) lie between the two largest
- 20 replicates of the whole sweep: 140 fits, 80 per size and parameter
- All fits sampled well: $\hat{R} \leq 1.006$, bulk ESS $\geq$ 1,554

Model A on synthetic data: the 95% HDIs are calibrated



Model A on synthetic data: γ_l is pulled upward



True γ_l (0.005, 0.01) sits in the prior's lower tail, and the data hold only 3–7 such lapses: the estimates are 1 sd too high at small sizes, 0.36 sd at 16,000.

Model A on synthetic data: what is recoverable

At the size of the real data (about 3,230 trials per group):

- **Recoverable:** PSE, JND, β_0, γ_h: unbiased, calibrated HDIs
  - posterior sd about 0.3 μm for the PSE, 0.2 μm for the JND
- **Recoverable, HDIs a little wide:** β_1 (coverage 0.98–0.99)
- **Not reliably recoverable:** small γ_l
  - biased upward by 0.4–0.5 sd; by about 1 sd when the curve never reaches its ceiling (left_uni), where γ_l trades off with the slope
  - HDIs still cover the truth, but the posterior mean should not be read literally
- **Caveats:** one set of true values; data simulated with one curve per group, so this says nothing about variation between sessions

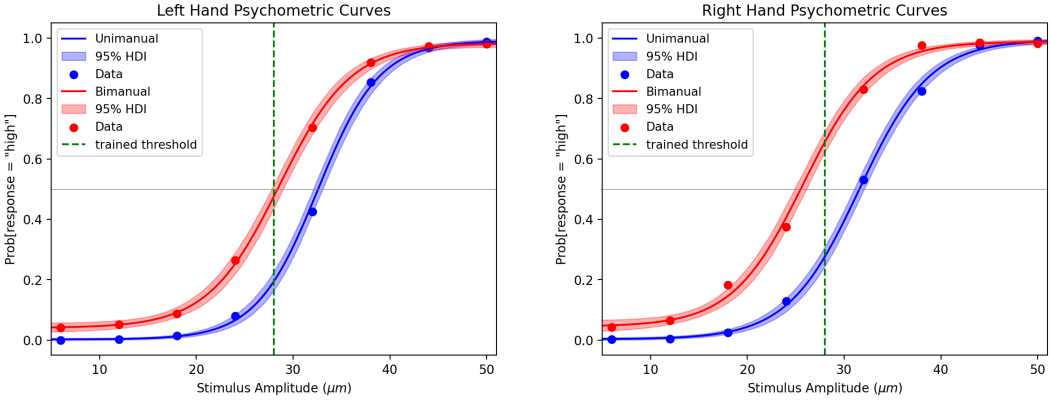
Model A on the real data: group estimates

	PSE (μm)	JND (μm)	γ_h	γ_l
left unimanual	32.58 [32.10, 33.04]	3.56 [3.25, 3.89]	0.002 [0.000, 0.006]	0.009 [0.000, 0.018]
left bimanual	28.38 [27.79, 28.98]	4.39 [3.93, 4.83]	0.041 [0.025, 0.056]	0.016 [0.005, 0.027]
right unimanual	31.57 [31.05, 32.06]	4.09 [3.77, 4.41]	0.003 [0.000, 0.008]	0.006 [0.000, 0.013]
right bimanual	25.50 [24.89, 26.08]	4.36 [3.93, 4.81]	0.045 [0.025, 0.064]	0.012 [0.004, 0.020]

Posterior mean [95% HDI]. 12,937 trials, 4 chains $\times$ 5,000 draws: no divergences, $\hat{R} = 1.000$, bulk ESS $\geq$ 15,213.

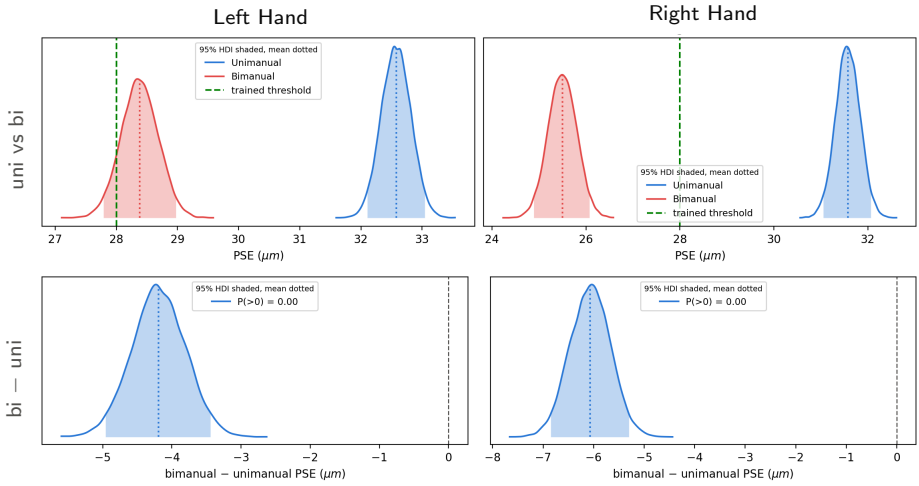
- Without the distractor the PSE lies 3.6–4.6 μm above the trained 28 μm
- With it the PSE moves to the threshold (left) or below it (right)
- γ_l estimates lean high (see the synthetic study)

Model A on the real data: fitted curves



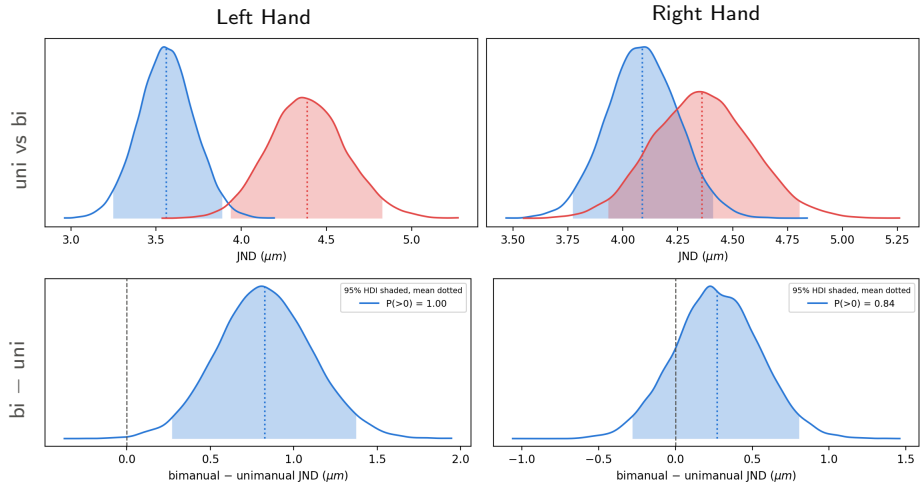
Unimanual in blue, bimanual in red, with the observed proportions pooled over sessions; dashed: the trained threshold of 28 μm.

Model A on the real data: effect of the distractor on PSE

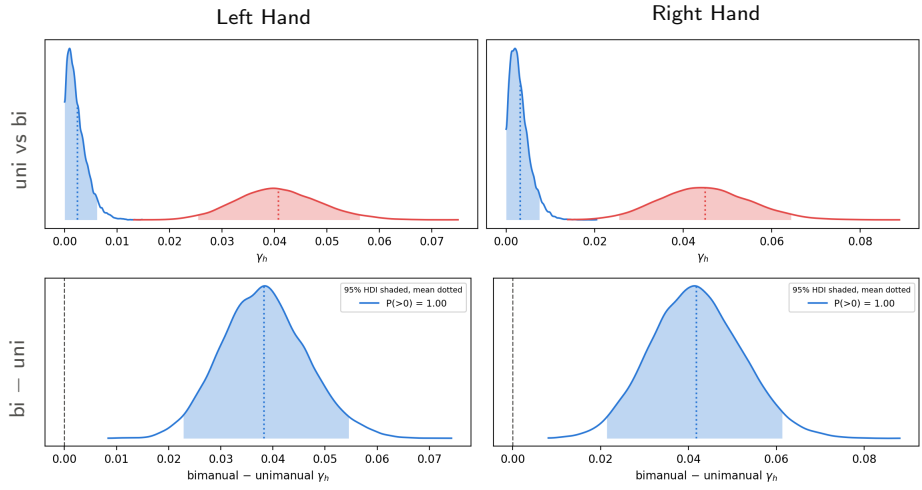


Unimanual Bimanual 95% HDI shaded, mean dotted; green dashed: trained threshold

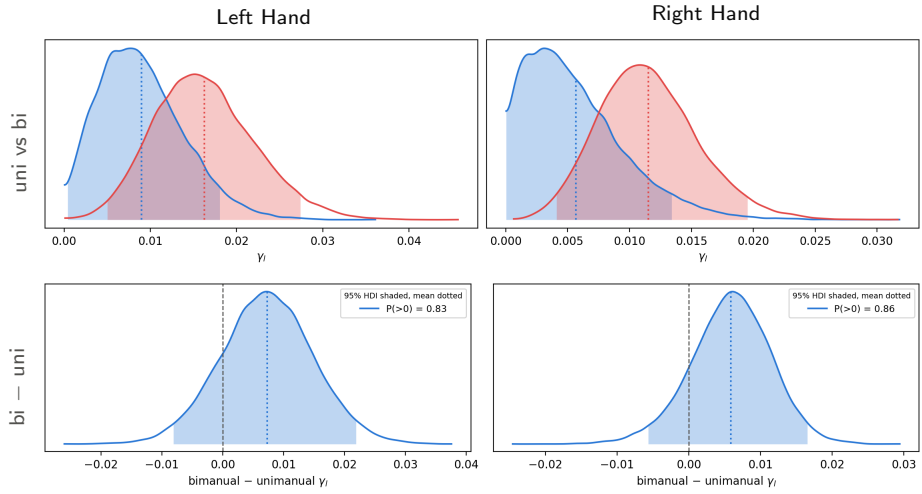
Model A on the real data: effect of the distractor on JND



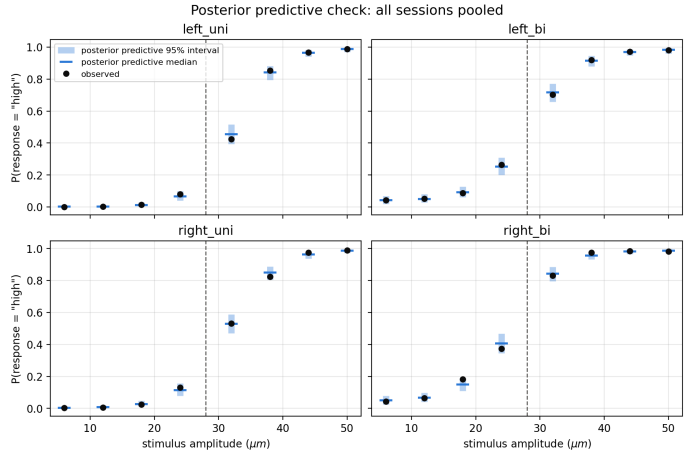
Model A on the real data: effect of the distractor on γ_h



Model A on the real data: effect of the distractor on γ_I



Model A on the real data: the pooled data are reproduced



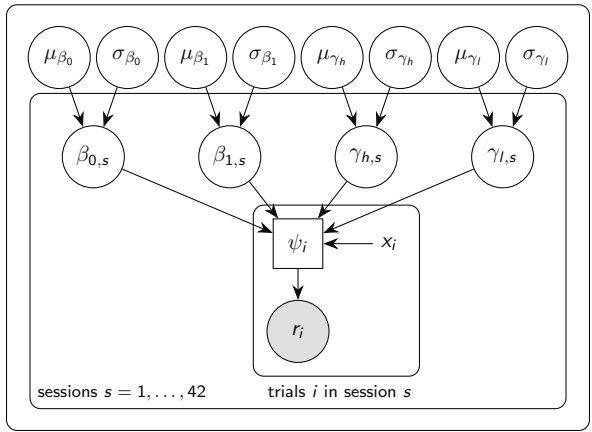
Pooled over sessions, all 32 group $\times$ amplitude proportions lie inside their 95% posterior predictive intervals.

Model A: conclusions

- Reliable fit; the pooled data are reproduced at every amplitude ($\text{elpd}_{\text{loo}} = -3344.0$, SE 64.5)
- Effect of the distractor (bimanual − unimanual), posterior mean [95% HDI]:
  - **PSE:** lowered, by -4.19 [-4.96, -3.44] μm (left) and -6.07 [-6.85, -5.30] μm (right)
  - **JND:** raised for the left hand, $+0.83$ [0.27, 1.37] μm; unclear for the right, $+0.27$ [-0.29, 0.80] μm
  - γ_h**:** raised, by $+0.038$ [0.023, 0.055] (left) and $+0.042$ [0.021, 0.061] (right)
  - γ_l**:** no clear effect, $+0.007$ [-0.008, 0.022] (left) and $+0.006$ [-0.006, 0.017] (right)

$\Rightarrow$ A curve for every session: Model B

Model B: graphical model



Circles: random variables (shaded: observed); rectangle: deterministic; x_i: fixed covariate. The same model is fit independently to each group.

Model B: likelihood

For each group (left/right $\times$ unimanual/bimanual), fit independently to its $n \approx 3{,}230$ trials, now with a parameter set for every session s:

Likelihood: probability of obtaining responses $\vec{r}$ given parameters $\vec{\theta} = (\vec{\theta}_s)_{s=1}^{42}$,
$\vec{\theta}_s = (\beta_{0,s}, \beta_{1,s}, \gamma_{h,s}, \gamma_{l,s})$
$r_i \sim \textbf{Bernoulli}[\psi(x_i; \vec{\theta}_{s[i]})]$ ($s[i]$: the session of trial i), so:

$$\mathbf{P}[r_i = z] = \psi(x_i; \vec{\theta}_{s[i]})^z \left(1 - \psi(x_i; \vec{\theta}_{s[i]})\right)^{1-z}, \qquad z \in \{0, 1\}$$

$$\begin{aligned} L(\vec{\theta}; \vec{r}) &= \prod_{i=1}^{n} \mathbf{P}[r_i \mid \vec{\theta}_{s[i]}] \\ &= \prod_{s=1}^{42} \prod_{i:\, s[i]=s} \psi(x_i; \vec{\theta}_s)^{r_i} \left(1 - \psi(x_i; \vec{\theta}_s)\right)^{1-r_i} \end{aligned}$$

Model B: priors

Session parameters drawn from the group's population (non-centered in the code):

$$\beta_{0,s} \sim \mathcal{N}(\mu_{\beta_0},\ \sigma_{\beta_0}), \qquad \gamma_{h,s} = 0.25\, w_{h,s}, \quad w_{h,s} \sim \text{LogitNormal}(\mu_{\gamma_h},\ \sigma_{\gamma_h})$$
$$\beta_{1,s} \sim \mathcal{N}(\mu_{\beta_1},\ \sigma_{\beta_1}), \qquad \gamma_{l,s} = 0.25\, w_{l,s}, \quad w_{l,s} \sim \text{LogitNormal}(\mu_{\gamma_l},\ \sigma_{\gamma_l})$$

Hyperpriors, the same in every group:

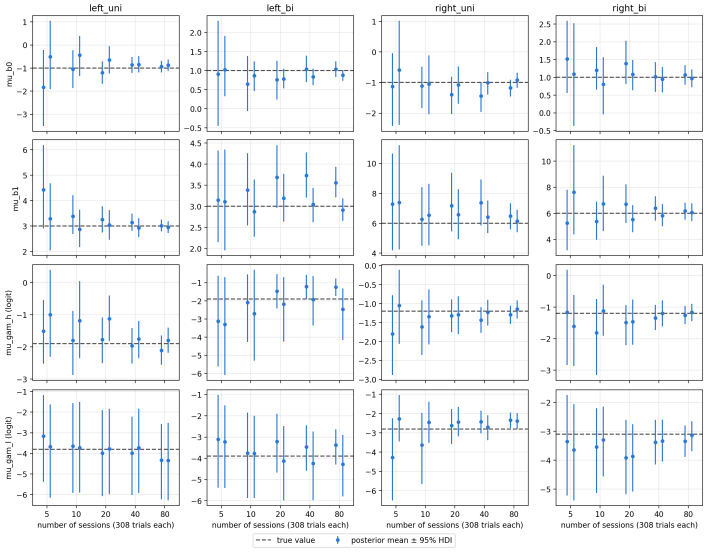
$$\mu_{\beta_0} \sim \mathcal{N}(0,\ 2), \qquad \sigma_{\beta_0} \sim \text{HalfNormal}(2),$$
$$\mu_{\beta_1} \sim \text{LogNormal}(\log 4,\ 0.5), \qquad \sigma_{\beta_1} \sim \text{Exponential}(1),$$
$$\mu_{\gamma_h},\ \mu_{\gamma_l} \sim \mathcal{N}(-3,\ 1.5), \qquad \sigma_{\gamma_h},\ \sigma_{\gamma_l} \sim \text{HalfNormal}(0.5)$$

- Chosen from the design, checked by prior predictive simulation, not fit to the data
- μ_{β_0}: typical PSE within 13–43 μm with 90% probability
- $\mu_{\beta_1} > 0$: median 4, 95% in [1.5, 10.6]; σ_{β_1} keeps 98% of session slopes positive
- Typical lapse rate: median 0.012, 95% in [0.0007, 0.12]

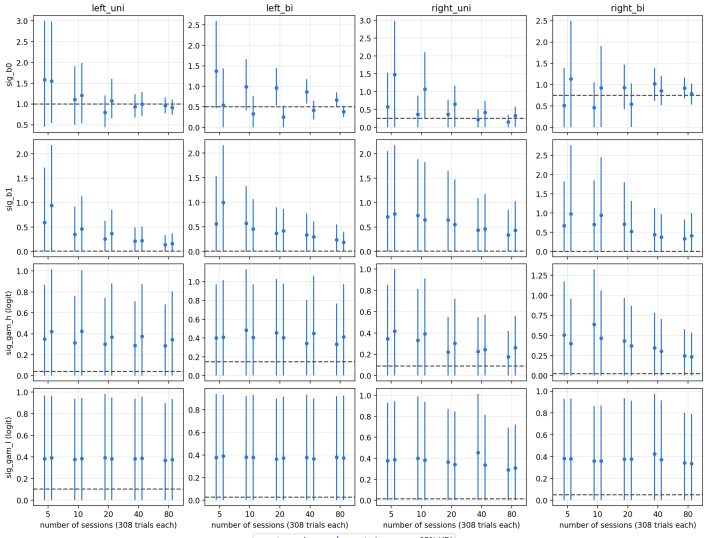
Model B on synthetic data: the study

- Fixed true hyperparameters per group (group slopes $\mu_{\beta_1} = 3$ for the left hand, 6 for the right); session parameters drawn from them
- Two sweeps, each run twice with new data:
  - sessions: 5, 10, 20, 40, 80 sessions of 308 trials (tests the hyperparameters)
  - trials: 42 sessions of 20, 40, 80, 160, 320 trials (tests the session parameters)
- The real data (42 sessions of about 308 trials) are closest to the 40-session and 320-trial fits
- All 20 fits sampled well: no divergences, $\hat{R} \leq 1.010$, bulk ESS ≥ 579

Model B on synthetic data: group means



Model B on synthetic data: between-session sds



Model B on synthetic data: session-level recovery

At 320 trials per session (42 sessions, both replicates): RMSE of the posterior mean / mean 95% HDI width / coverage

	left unimanual	left bimanual	right unimanual	right bimanual
PSE	0.14 / 0.49 / 0.90	0.12 / 0.42 / 0.90	0.05 / 0.16 / 0.93	0.09 / 0.32 / 0.94
JND	0.014 / 0.15 / 1.00	0.034 / 0.17 / 1.00	0.013 / 0.08 / 1.00	0.010 / 0.10 / 1.00
β_0	0.41 / 1.67 / 0.93	0.34 / 1.20 / 0.92	0.26 / 1.01 / 0.95	0.55 / 1.99 / 0.94
β_1	0.19 / 1.39 / 1.00	0.23 / 1.18 / 1.00	0.29 / 2.11 / 1.00	0.35 / 2.97 / 1.00
γ_h	0.003 / 0.047 / 1.00	0.012 / 0.052 / 1.00	0.009 / 0.071 / 1.00	0.013 / 0.062 / 1.00
γ_l	0.016 / 0.048 / 1.00	0.003 / 0.019 / 1.00	0.004 / 0.021 / 1.00	0.003 / 0.023 / 1.00

PSE and JND in standardized units (1 unit = 14.6 μm).

Model B on synthetic data: what is recoverable

At the size of the real data (42 sessions of about 308 trials):

- **Recoverable:** per-session PSE and JND; the group means μ_{β_0}; μ_{β_1} for the left hand; the bias spread σ_{β_0}; μ_{γ_h} except for left bimanual
- **Weakly constrained:** μ_{β_1} for steep curves (slope 6, right hand): the curve rises within one stimulus spacing, so only 2 levels inform it
- **Biased in some datasets:** a lapse rate whose side of the curve never reaches its asymptote: γ_l for left unimanual (too high), γ_h for left bimanual (too low)
- **Not identified:** σ_{β_1}, σ_{γ_h}, σ_{γ_l} when their true values are near 0; their posteriors reflect the prior
- Session-level HDIs mostly conservative; PSE coverage slightly low (0.90–0.94)

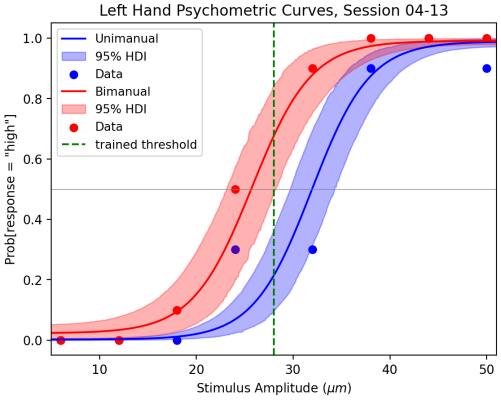
Model B on the real data: a typical session

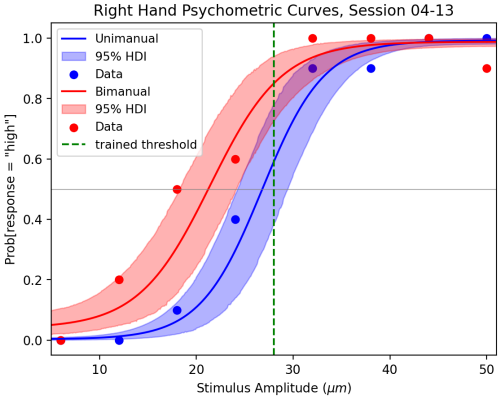
	PSE (μm)	JND (μm)	γ_h	γ_l
left unimanual	32.4 [31.6, 33.3]	3.29 [2.95, 3.63]	0.002 [0.000, 0.005]	0.009 [0.000, 0.019]
left bimanual	28.2 [27.2, 29.2]	3.89 [3.34, 4.46]	0.023 [0.009, 0.037]	0.009 [0.001, 0.017]
right unimanual	31.4 [30.3, 32.5]	3.56 [3.23, 3.89]	0.003 [0.000, 0.006]	0.007 [0.000, 0.015]
right bimanual	25.5 [24.5, 26.5]	3.95 [3.48, 4.44]	0.036 [0.017, 0.056]	0.010 [0.003, 0.018]

Quantities at the group-level means, posterior mean [95% HDI]. 4 chains $\times$ 5,000 draws: no divergences; $\hat{R} = 1.00$ and bulk ESS $\geq$ 2,210 for all but 4 raw variables, whose only disagreement is a sign that does not enter the model.

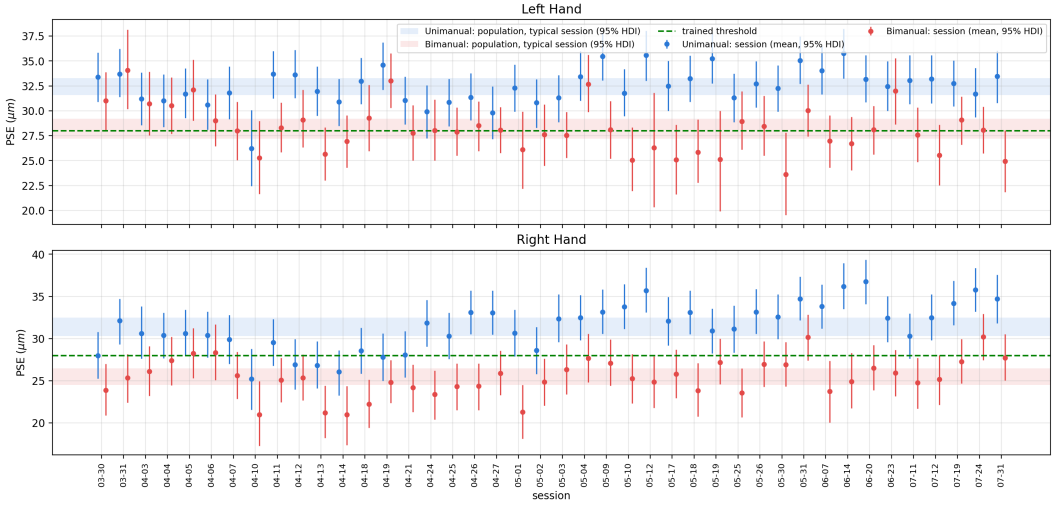
- The sessions differ in bias ($\sigma_{\beta_0} = 0.76$–1.01); left bimanual also in slope ($\sigma_{\beta_1} = 1.27$)
- The between-session lapse sds sit near their prior (not identified)

Model B on the real data: fitted curves in one session (04-13)

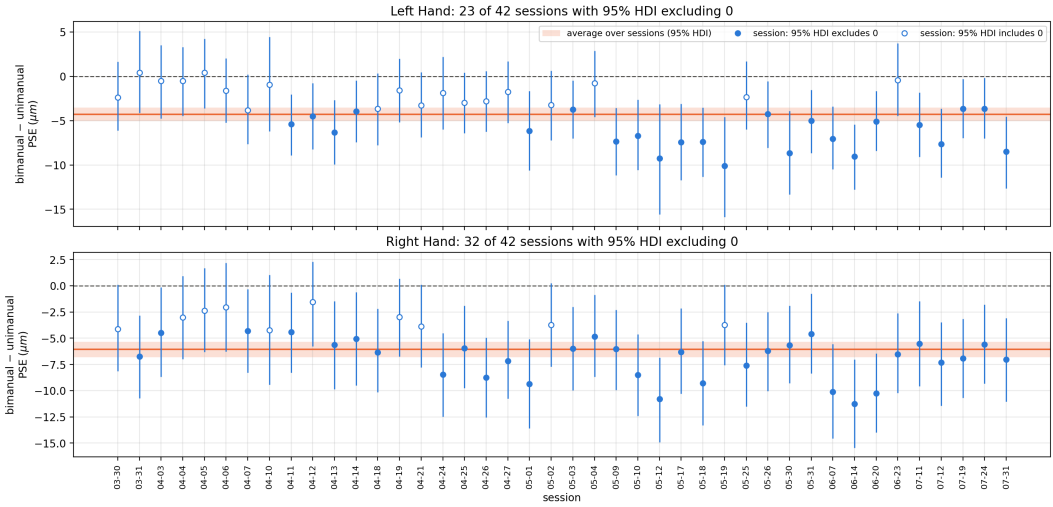




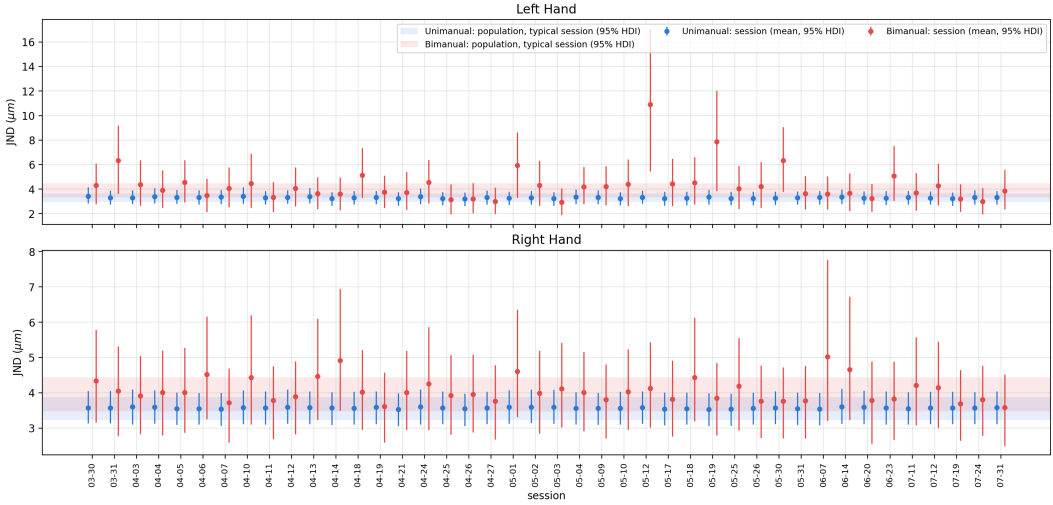
Model B on the real data: PSE in each session



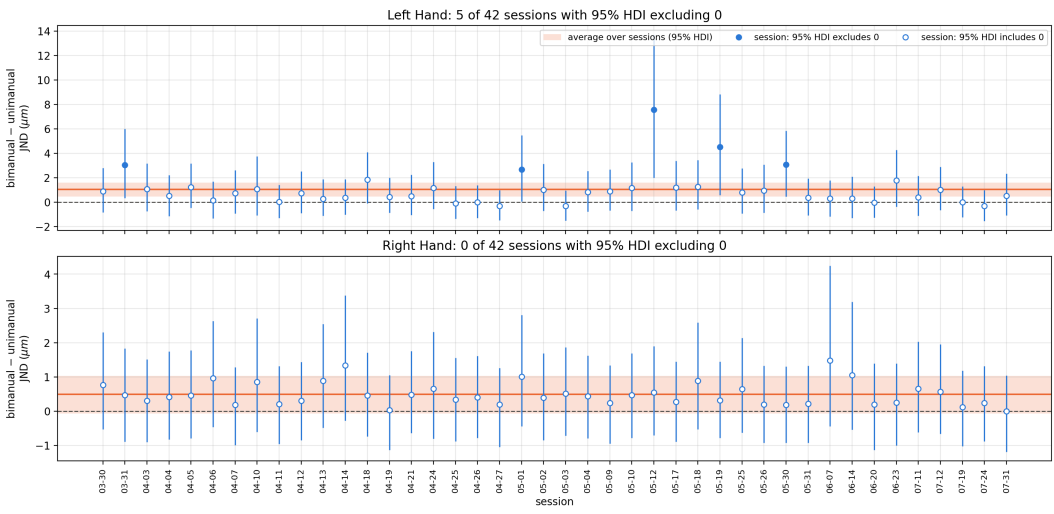
Model B on the real data: effect of the distractor on PSE



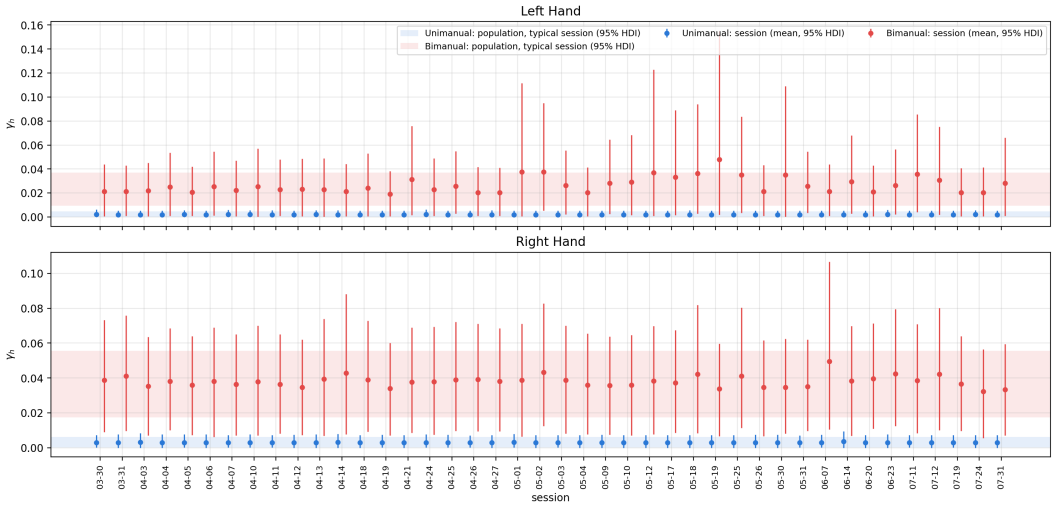
Model B on the real data: JND in each session



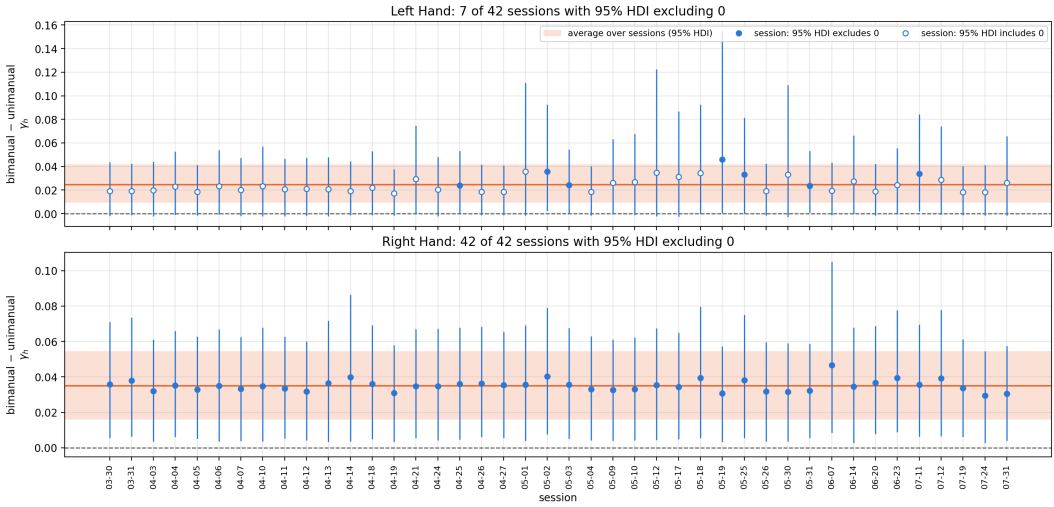
Model B on the real data: effect of the distractor on JND



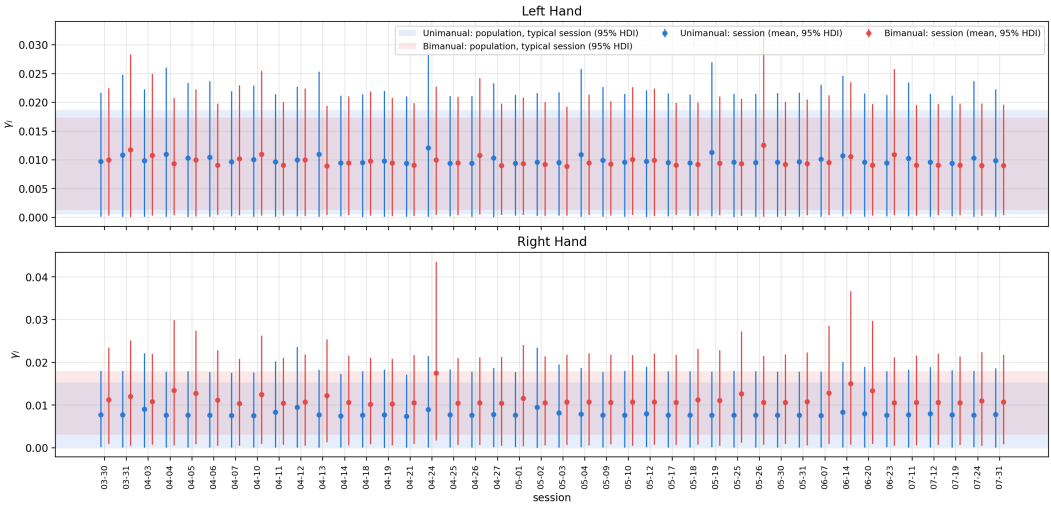
Model B on the real data: γ_h in each session



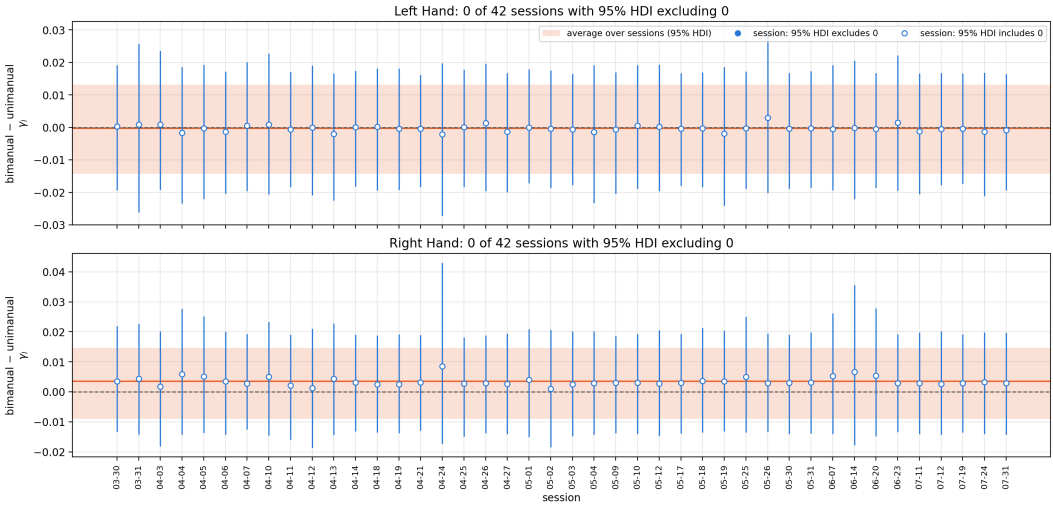
Model B on the real data: effect of the distractor on γ_h



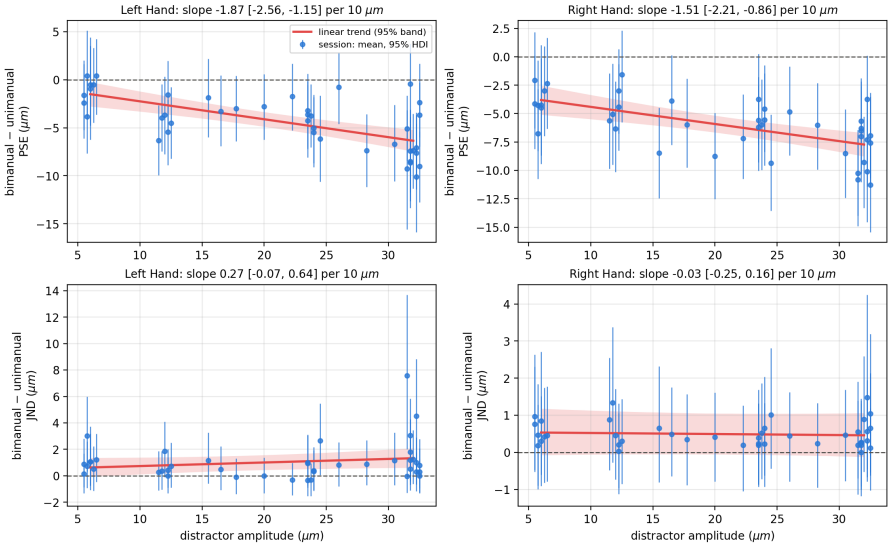
Model B on the real data: γ_l in each session



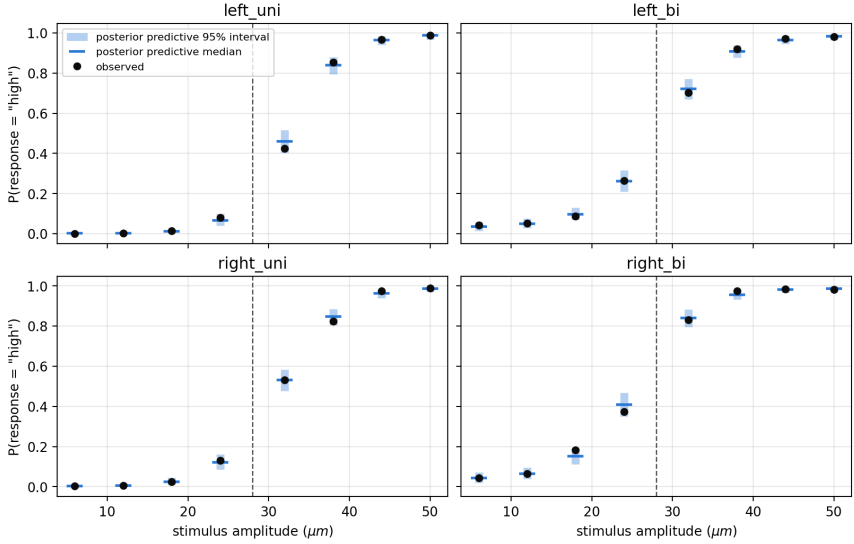
Model B on the real data: effect of the distractor on γ_l



Model B: PSE shift vs distractor amplitude (exploratory)



Model B on the real data: the pooled data are reproduced



Model B: conclusions

- Reliable fit; no systematic misfit, pooled or within sessions ($\text{elpd}_{\text{loo}} = -3180.0$, SE 64.3)
- Effect of the distractor (bimanual − unimanual), average over the 42 sessions, posterior mean [95% HDI]:
  - **PSE:** lowered, by -4.29 [-5.05, -3.52] μm (left) and -6.06 [-6.80, -5.34] μm (right); negative in 40/42 and 42/42 sessions
  - **JND:** raised for the left hand, $+1.03$ [0.46, 1.59] μm, driven by a few sessions; unclear for the right, $+0.49$ [-0.07, 1.03] μm
  - γ_h**:** raised, by $+0.025$ [0.009, 0.042] (left) and $+0.035$ [0.016, 0.054] (right)
  - γ_l**:** no effect, 0.000 [-0.014, 0.013] (left) and $+0.003$ [-0.009, 0.015] (right)
- The PSE shift grows with distractor amplitude, by about -2 μm per 10 μm (exploratory; amplitude is confounded with session order)

$\Rightarrow$ Put the distractor amplitude into the model: Model C

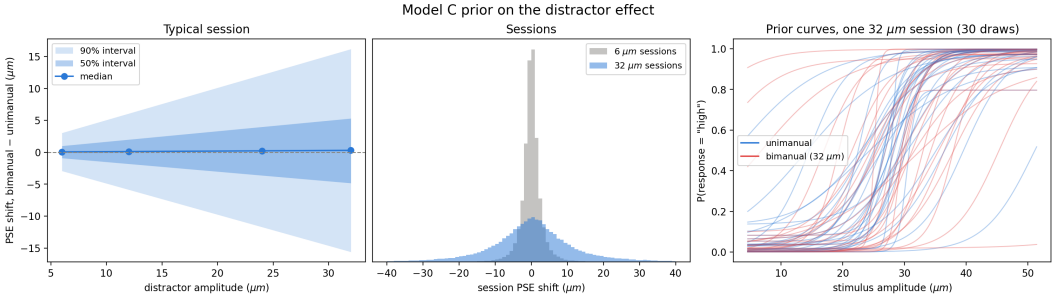
Model C

$$\psi(x, d) = \gamma_h + (1 - \gamma_h - \gamma_l)\ \text{logit}^{-1}(\beta_0 + \beta_1 x + \beta_2 d)$$

- d: distractor amplitude (0 on unimanual trials); one group per hand.
- Uni- and bimanual trials share slope and lapse rates: the distractor only shifts the curve, by an amount proportional to d.

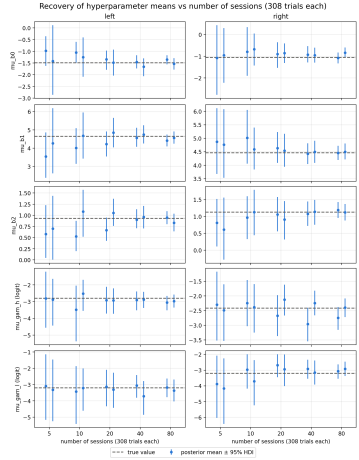
TODO: why: test the amplitude trend from Model B inside a model

Model C: prior on the distractor effect



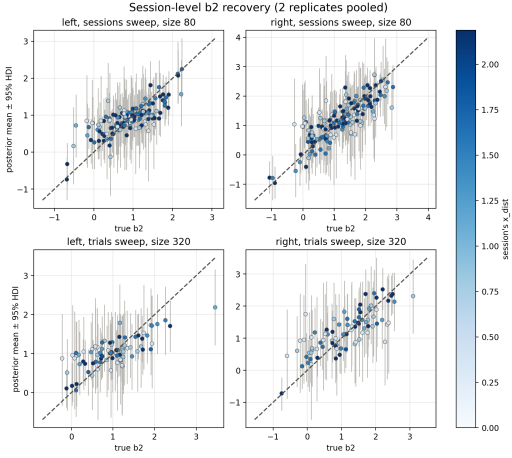
TODO: weakly informative; a weak distractor can only shift a little (Convergence_C_Report)

Model C: convergence on synthetic data



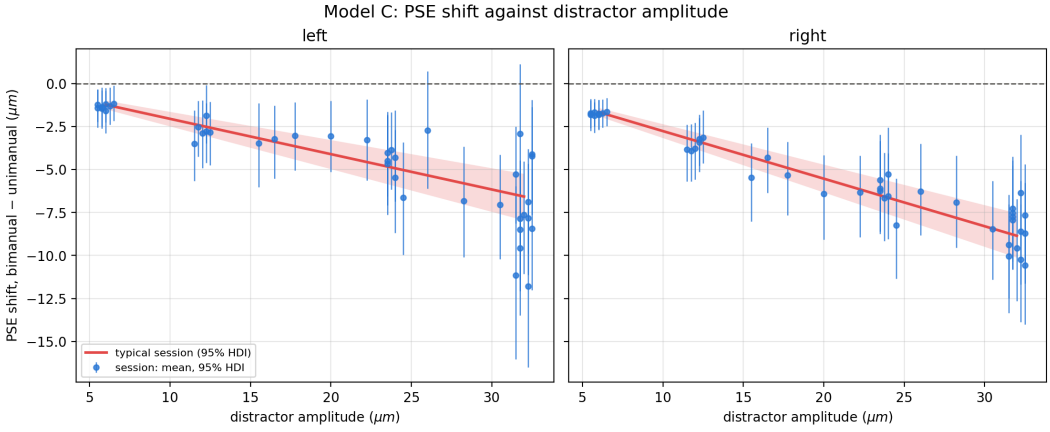
TODO: all hand-level means recovered, μ_{β_2} well (Convergence_C_Report, "Conclusions")

Model C: per-session β_2 is weak for small distractors



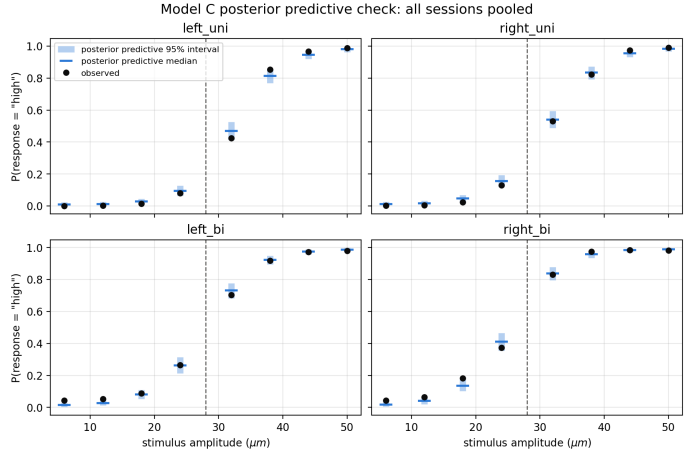
TODO: shrinkage toward the hand mean at 6 μm

Model C: the shift grows with distractor amplitude



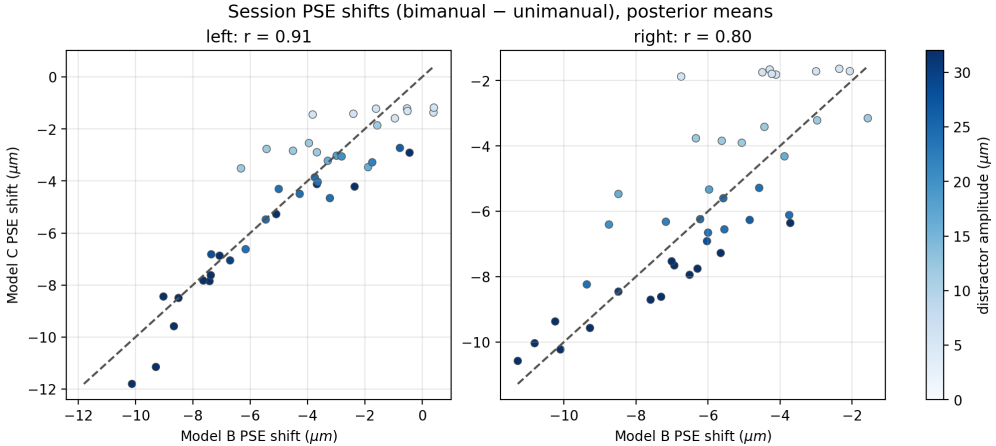
TODO: μ_{β_2} and the typical shift at 6 and 32 μm (Fit_C_Report, "Effect of the Distractor")

Model C misfits at the weakest stimuli



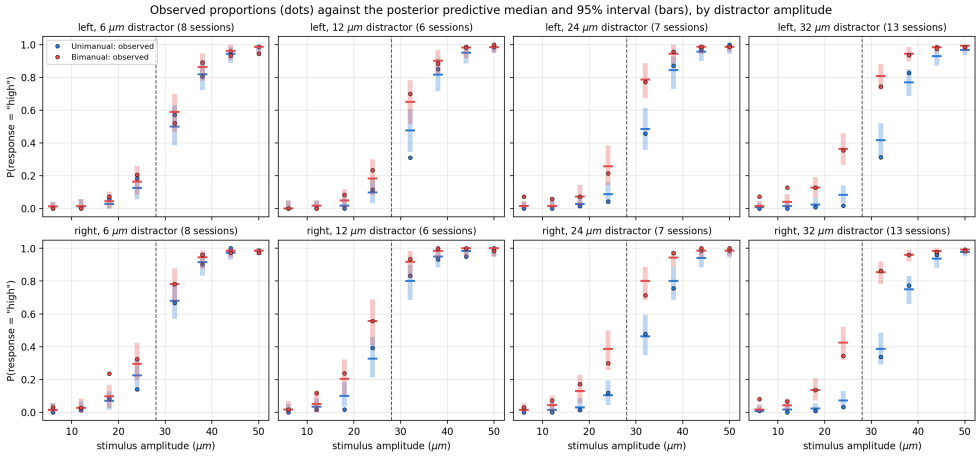
TODO: shared lapse rates: 6 of 32 proportions outside (Fit_C_Report, "Posterior predictive check")

Model C against Model B



TODO: LOO favors B by 37 (SE 12); 6 μm shifts too small (Fit_C_Report, Comparison)

Model C: shift by distractor amplitude, observed vs predicted



TODO: right hand at 6 μm outside: a presence effect (Fit_C_Report, amplitude check)

Comparison of the models

	A	B	C
Sessions	pooled	hierarchical	hierarchical
Distractor	separate groups	separate groups	covariate $\beta_2 d$
elpd_{loo}	**TODO:**	**TODO:**	**TODO:**
PSE shift, left / right (μm)	**TODO:**	**TODO:**	**TODO:**

TODO: fill from Fit_A/B/C_Report; elpd differences from Fit_B and Fit_C

Conclusions

- The distractor lowers the PSE: **TODO:** size, both hands.
- It raises the rate of "high" responses to weak stimuli (γ_h).
- The sessions differ; a hierarchical model is needed.
- The shift grows with distractor amplitude, **TODO:** but not proportionally?

Next steps

- Revise Model C: lapse rates per condition, a distractor-presence term, possibly a bimanual slope.
- Model recovery for B against the revised C.
- **TODO:** further data or experiments